University of Bamberg

Master Project - Machine Listening
(CH-Proj-M)
02 – Scientific Writing

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Lecture
Structure

Week	Day	Project Phase	Topics
1	13.04.		Overview, master project organization, open project topics
2	20.04.	**1. Literature research**	Literature research strategies, reference management
3	27.04.		Basics: academic writing, required structure of the term paper
4	04.05. **(P1)**	**2. Data acquisition** (Dataset research, data recording)	Role of data and annotations in machine listening research, data protection, licences (CC etc.)
5	11.5.		Basics: audio editing and annotation (Audacity, Sonic Visualizer)

Scientific Writing
Motivation

- Crucial skill for researchers to effectively communicate their research results
- Well-written papers enhance the credibility and the impact of research
- High quality publications
  - Researchers are evaluated based on writing output (number of publications, publication type, journal rank etc.)
  - Essential for research funding & academic career

Scientific Writing
Motivation

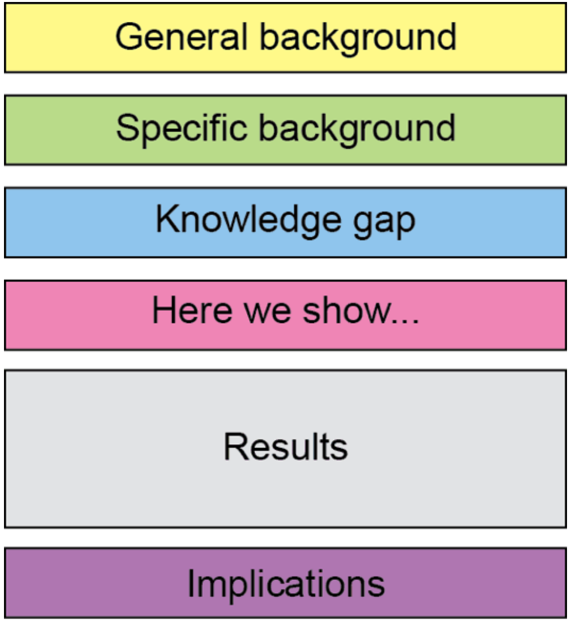
- Papers follow a common structure that helps readers navigate the paper and understand its contents efficiently
- Following this “recipe” improves the likelihood of getting published
  - Reviewers know what to expect
- Scientific writing is a skill which takes years of practice

Scientific Writing
Title

- The title should be concise and informative, accurately reflecting the content of the paper
- As short as possible, as long as necessary
- Use relevant/common keywords from your field
  - capture the essence of your study
  - helps in search engine optimization

Scientific Writing
Abstract

- Brief summary of the paper
  - objectives, methods, results, and conclusions
- Rule-of-thumb: one sentence per section
- Follow required max. word count



https://mitcommlab.mit.edu/be/wp-content/uploads/sites/2/2016/08/journal-abstract1.gif

Scientific Writing
Introduction

- Introducing context / background of research (e.g. overarching research field)
- Introduce and motivate research problem/objective (why is it interesting / useful?), possible application scenarios
- Mention challenges (why is it hard?)
- Summarize main contribution(s) – summarize significance of the publication
- Optional: brief overview over paper structure

Scientific Writing
Related Work

- Compare: provide overview over (relevant) related work
  - Summarize main findings / approaches
  - Group references based on similar methodologies, theories, or findings

Scientific Writing
Related Work

- Contrast: Highlight how your research differs from the existing literature
  - e.g.: different methodologies, new data, alternative interpretations, or addressed research gaps
- State how your research adds to the existing body of knowledge
  - new insights, extending current understanding, or refining existing models or theories

Scientific Writing
Methodology

- Describe the methods and techniques used to conduct the research
  - Add references for established methods
  - Possible use of equations, flowcharts, figures
- Possible subsections
  - Feature extraction
  - Neural network architecture

Scientific Writing

Evaluation & Results

- Describe experimental design
  - Hyperparameter configurations which are evaluated
  - Baseline system
  - Evaluation metrics
  - Dataset(s) (possible subsection)
- Summarize results
  - Use tables, figures, and graphs to illustrate key findings and trends

Scientific Writing
Conclusions

- Repeat main research objectives
- Summarize the key findings of the study and their significance
- Interpret the results and places them in the broader context of existing knowledge
- (Optional) propose recommendations for further research or practice

Scientific Writing

Ethical use of AI (a suggestion)

Deep thoughts, creative, research skills (manual)

↓ *Manual or AI possible*

Use AI (debug, review, refine)

Topic/Task/Research field

↓ *Identify relevant related work / datasets*

Literature / dataset research → **Practical part (experiments)** → **Scientific writing** → **Manuscript**

Literature / dataset research

- *Read*
- *Summarize*
- *compare/contrast*

Practical part (experiments)

- *Plan experiments*
- *Implement* and *debug*
- *Summarize/visualize results*

Scientific writing

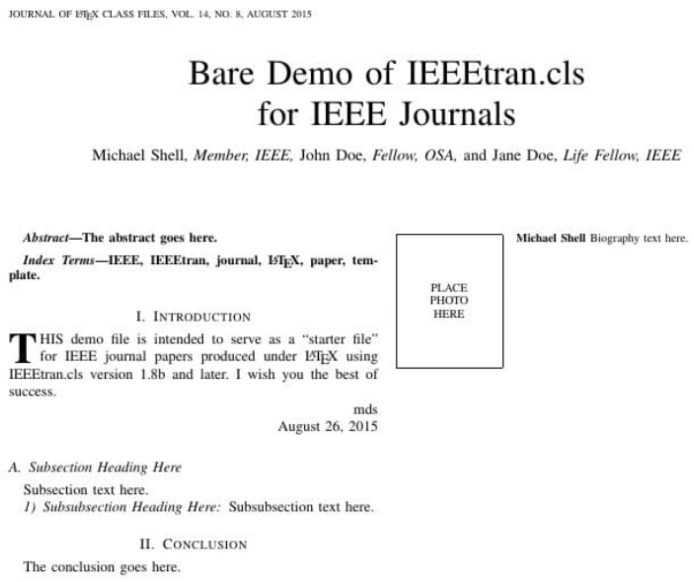
- *Conceptualize*
- *Write first draft*
- *Improve (iterate)*
- *Polish (language / style, references)*

Manuscript

- *Use AI as critical reviewer*

Term Paper
Required Structure

- IEEE two-column conference paper
  - Overleaf template
- 6 pages, PDF format, English
- Code repository link
- 8-10 references
- Structure as discussed before
- AI Transparency Statement
  - Tools, prompt documentation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Bare Demo of IEEEtran.cls for IEEE Journals

Michael Shell, *Member, IEEE,* John Doe, *Fellow, OSA,* and Jane Doe, *Life Fellow, IEEE*

***Abstract*—The abstract goes here.**

***Index Terms*—IEEE, IEEEtran, journal, LATEX, paper, template.**

I. Introduction

THIS demo file is intended to serve as a "starter file" for IEEE journal papers produced under LATEX using IEEEtran.cls version 1.8b and later. I wish you the best of success.

mds
August 26, 2015

A. Subsection Heading Here

Subsection text here.

1) Subsubsection Heading Here: Subsubsection text here.

II. Conclusion

The conclusion goes here.

PLACE PHOTO HERE

Michael Shell Biography text here.

Thank you for your attention!
Any questions?

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